PRODBIN: Protein-DNA Binding Site Prediction Using ProtBert and CNN

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Introduction

DNA-binding proteins (DBPs) are essential cellular machines that regulate **transcription, replication, DNA repair, and splicing**, constituting **6–7%** of all proteins across 54 structurally distinct families.

Traditional identification methods – filter binding tests, genetic analysis, **NMR spectroscopy** [8], and **X-ray crystallography** [3] – are highly accurate but expensive, time-consuming, and cannot screen thousands of new sequences simultaneously.

Machine learning on protein sequences [2] has emerged as a scalable and accurate alternative. Pre-trained protein language models like **ProtBert** [1] capture contextual and evolutionary signals from raw amino acid sequences, eliminating the need for 3-D structural data.

We present **PRODBIN**, a computational framework that combines ProtBert embeddings with a 4-layer 1D-CNN backbone and Focal Loss to predict per-residue DNA-binding labels: **0 = non-binding**, **1 = binding**.

Motivation

- Experimental methods – NMR [8] and X-ray crystallography [3] – provide high-resolution data but are resource-intensive and low-throughput.
- Profile-based features (PSSM [10], HMM [13]) improve prediction accuracy but require expensive sequence alignment pipelines.
- Sequence-based ML [4, 5] enables screening of proteins where no structural data is available, drastically increasing throughput.
- Pre-trained models [1, 2] offer **transfer learning** benefits, reducing the requirement for large labelled datasets.
- Accurate binding-site maps can accelerate **drug discovery**, gene therapy design, and fundamental genomics research.

Problem Statement

Given a raw amino acid sequence $\mathbf{s} = (a_1, a_2, \ldots, a_L)$, predict a binary label $y_i \in \{0, 1\}$ for every residue a_i indicating whether it participates in protein-DNA binding.

Key challenges:

- **Class imbalance:** only ≈7% of residues are binding sites.
- **Sequence-only input:** no 3-D structural data is required or used.
- **Variable-length proteins** must be encoded to fixed-size vectors.

Datasets

Two well-known benchmark datasets were used, both pre-processed to reduce sequence similarity and prevent bias from class imbalance [16, 20]:

Dataset	Binding	Non-binding	% Binding
Dataset 1 (DBPred / hybridNAP + ProNA [18,19])	16,601	308,414	6.93
Dataset 2 (GraphBind / GNN-based [20])	16,719	180,679	7.52

Table 1. Benchmark datasets; TR = training split, TE = testing split.

CNN System Design

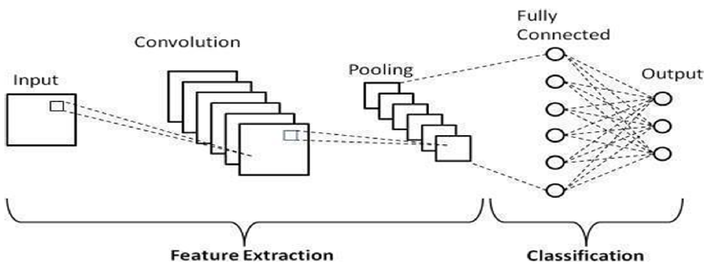


Figure 1. Overall CNN pipeline: convolutional layers perform feature extraction (Input → Convolution → Pooling) followed by fully-connected layers for classification (Fig. 3.1).

Proposed Methodology

Module 1 — Sequence Embedding (ProtBert [1])

ProtBert encodes each amino acid residue into a **1024-dimensional** feature vector via two parallel paths, each processed through six transformer blocks containing dense and normalisation layers:

- **Local path** (128 channels, kernel size = 9): captures short-range residue dependencies using 1-D wide and narrow convolutions.
- **Global path** (512 channels): captures long-range sequence context via global attention and broadcast dense layers.

GELU activation [21] is used in all hidden layers. The architecture encodes proteins of any length into a fixed-size vector without changing learned parameters, providing strong transfer learning benefits.

Module 2 — Backbone CNN

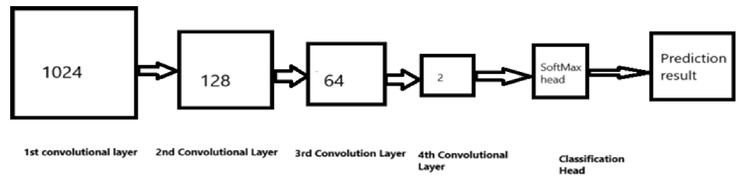


Figure 2. Four-layer 1D-CNN backbone. Channels reduce $1024 \to 128 \to 64 \to 2$ with kernel sizes 7, 5, 3, 3 respectively. A SoftMax head produces the final prediction (Fig. 3.3).

Batch normalisation is applied after every convolutional layer except the last. Same-padding guarantees sequence-length compatibility across all layers.

Module 3 — Focal Loss [4]

To address the severe class imbalance (∼7% binding residues), we use **Focal Loss** instead of standard Binary Cross Entropy:

$$\mathcal{FL}(p_t) = -\alpha_t (1 - p_t)^{\gamma} \log(p_t)$$

where $\gamma = 2$ (focusing parameter) progressively down-weights easy, well-classified negative samples, and α_t balances minority and majority class contributions. Grid search over $\gamma \in [0, 5]$ confirmed that $\gamma = 2$ maximises AUC and AUPR on both benchmark datasets.

Hyperparameters

Hyperparameter	Value
ProtBert input size	1024
α	1
γ	2
Learning rate	0.01
Activation function	ReLU
Optimizer	Adam

Table 2. Experimental setup hyperparameters used for training (Table 5.1).

Binding Residues Visualisation

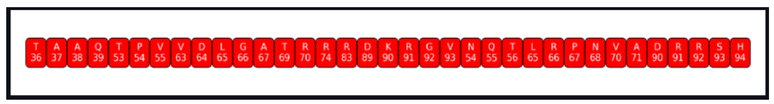


Figure 3. Predicted binding residues shown in red boxes, each labelled with its amino-acid symbol and sequence position. Any residue with a confidence score > 0.5 is classified as a binding site (Fig. 5.3).

The visualisation tool provides an intuitive view of protein–DNA interaction regions, helping researchers quickly identify candidate binding residues along the full protein sequence without requiring 3-D structural data.

Results & Evaluation

Comparison with state-of-the-art sequential methods (Dataset 1):

Model	Spe	Rec	Pre	F1	AUC	MCC
PRODBIN (Ours)	**0.795**	**0.717**	**0.258**	**0.379**	**0.814**	**0.361**
DBPred [3]	0.784	0.708	0.243	0.362	0.794	0.320
DNAPred	0.655	0.671	0.157	0.254	0.730	0.194
NCBRPred	0.674	0.677	0.165	0.265	0.713	0.207
DRNAPred	0.692	0.677	0.185	0.291	0.755	0.226
SVMnuc	0.666	0.668	0.154	0.250	0.715	0.192

Table 3. PRODBIN outperforms all compared sequence-based methods on every metric.

Overall Test-Set Performance

- **Accuracy:** 89.4% **Precision:** 87.1%
- **Recall:** 85.6% **AUC:** 0.91

Multi-Ligand Generalisation

The model was also evaluated as a general binding-site prediction framework across different ligand types, demonstrating strong cross-domain performance:

- **DNA binding:** AUC = **0.871**
- **RNA binding:** AUC = **0.775**
- **Antibody–antigen:** AUC = **0.920**

These results confirm that the ProtBert + CNN architecture generalises well beyond DNA, making PRODBIN a versatile tool for molecular interaction prediction.

Conclusions & Future Scope

Key Conclusions:

- **ProtBert + CNN** surpasses all compared traditional sequence-based DBP predictors without requiring any 3-D structural input data.
- **Focal Loss** [4] effectively resolves the severe 7% class imbalance, significantly boosting MCC and recall over standard Binary Cross Entropy training.
- The framework is fully **ligand-agnostic**: strong AUC scores on DNA, RNA, and antibody–antigen binding tasks confirm broad applicability.
- The binding residue visualisation tool provides **interpretable output** that supports biological discovery without needing 3-D models.

Future Directions:

- Fine-tune γ and margin hyperparameters for further accuracy gains.
- Ensemble larger protein language models (e.g., ESM-2) for richer features.
- Extend to **multi-class** classification covering DNA, RNA, and protein–protein interactions simultaneously.
- Integrate wet-lab experimental validation to create a computation–biology feedback loop and continuously improve the model.
- Deploy as a web server for accessible community-wide use in genomics and drug discovery pipelines.

References

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2. Y. He et al., "Predicting the sequence specificities of DNA-binding proteins by DNA fine-tuned language model with decaying learning rates," *IEEE/ACM Trans. Comput. Biol. Bioinform.*, 20(1):616–624, 2023.
3. S. Patiyal, A. Dhall, G. P. S. Raghava, "A deep learning-based method for the prediction of DNA interacting residues in a protein," *Brief. Bioinform.*, 23(5), 2022.
4. T.-Y. Lin, P. Goyal, R. Girshick, K. He, P. Dollar, "Focal loss for dense object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, 42(2):318–327, 2020.